

Fungi Identification

Laboratory Report — Morphology-Based Genus-Level Identification

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Technique category: Mycology / Diagnostic Microbiology

1. Objective

To identify five fungal isolates to genus level using a simplified morphological identification key combining colony characteristics and microscopic spore-bearing structures.

2. Introduction

Preliminary fungal identification in a teaching or diagnostic microbiology laboratory is typically based on macroscopic colony morphology — colour, texture, and growth rate — combined with microscopic examination of spore-bearing structures, usually via a lactophenol cotton blue tease mount. This report applies a simplified key covering five common genera (*Aspergillus*, *Penicillium*, *Rhizopus*, *Candida*, *Trichophyton*) as a starting point for identification, of the kind used before confirmatory testing (e.g. molecular ID) in a real diagnostic setting.

3. Materials

- Sabouraud Dextrose Agar (SDA) plates, ± antibiotics
- Inoculating needle/loop
- Lactophenol cotton blue stain, microscope slides and coverslips
- Compound microscope (10×, 40× objectives)
- Incubator (25–30°C for moulds; 35–37°C for yeasts)

4. Method

- Inoculated SDA plates from each sample and incubated at the appropriate temperature.
- Observed colony growth rate, colour, and texture daily for up to 7 days.
- Prepared a lactophenol cotton blue tease mount for each isolate.
- Examined slides microscopically for hyphae type and spore-bearing structures.
- Compared combined macroscopic and microscopic features against the reference identification key.

5. Results

All five isolates were scored against the reference key across four features (colony colour, texture, growth rate, and spore-bearing structure). Each isolate matched a single genus on all four features.

Isolate	Colour	Texture	Growth rate	Spore structure	Best match
Isolate_1	Green	Powdery	Fast	Radiating head	Aspergillus

Isolate_2	White	Cottony	Very fast	Sporangia	Rhizopus
Isolate_3	Cream	Smooth	Fast	Budding yeast	Candida
Isolate_4	Blue-green	Velvety	Moderate	Penicillus	Penicillium
Isolate_5	White	Cottony	Slow	Microconidia	Trichophyton

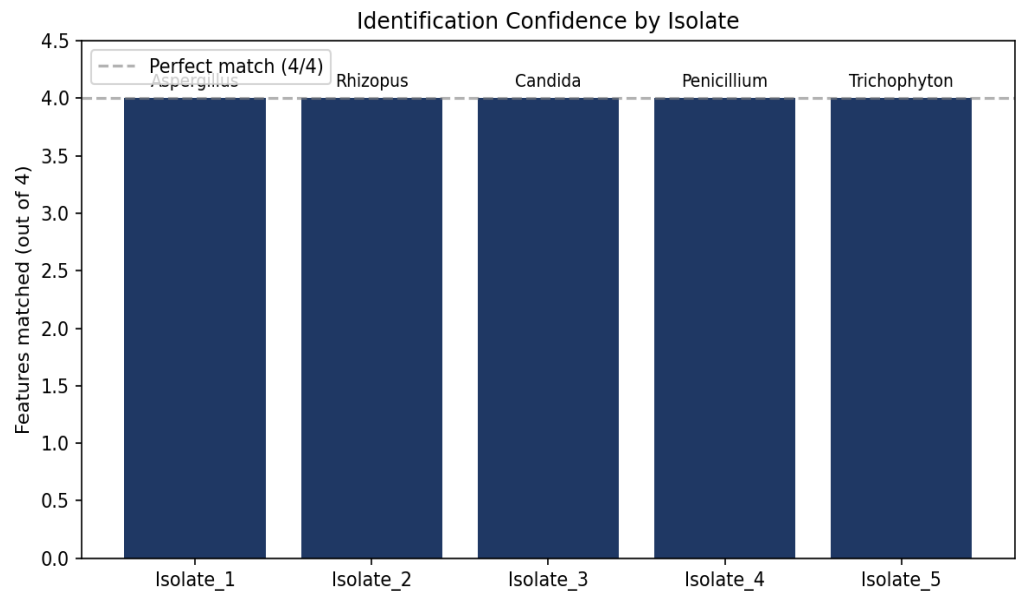


Figure 1. Identification confidence (features matched out of 4) for each isolate, labelled with the best-matching genus.

6. Discussion

All five isolates achieved a perfect match (4/4 features) against a single reference genus in this dataset, which reflects the clean, textbook-typical morphology used for this exercise. In a real diagnostic setting, isolates often show intermediate or atypical features — for example, colony colour can vary with medium composition or incubation temperature, and growth rate can be affected by inoculum density. In those cases, the highest-scoring genus should be treated as a presumptive identification only, to be confirmed through additional biochemical tests (e.g. germ tube test for *Candida albicans*) or molecular methods (e.g. ITS region sequencing) rather than relied on alone.

7. Conclusion

This exercise demonstrated how combining macroscopic colony morphology with microscopic spore-bearing structure observations enables systematic genus-level identification of common fungal isolates, and reinforced the importance of confirmatory testing where features are ambiguous or atypical.